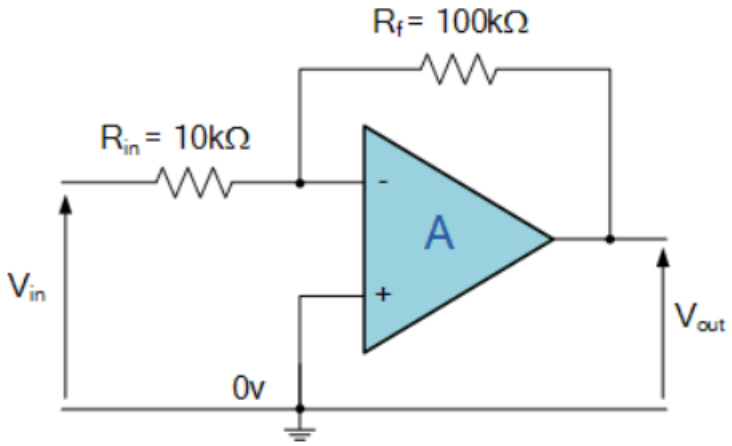
EE 3400 – Measurement and Instrumentation Tutorial.

Strain Gauges

1. What is the working principle of a strain gauge?
2. Define the Gauge factor of a strain gauge
3. What are the advantages of using a wheatstone bridge?
4. A strain gauge of 120Ω resistance has a gauge factor of 2.1. It is subjected to a strain of 500με. Calculate the change in resistance
5. In a wheatstone bridge, a strain gauge resistance increases from 120 Ω to 120.24 Ω due to strain. The other resistors are all 120 Ω. Calculate the output voltage if the input voltage is 6V

Amplifiers

1. What are the key assumptions made in the operation of operational amplifiers?
2. Explain what common-mode rejection ration is in a differential amplifier, and provide a formular for calculating it
3. What is an instrumentation amplifier? Draw a typical configuration of this amplifier and derive the formular for output voltage
4. Find the closed loop gain of the following inverting amplifier circuit.



5. If the gain of the amplifier above is increased to 40, find the value of the new resistors required